KIMQ8550 Ubuntu source code build guide

1. Preparation.....1
2. Get source code from HSPTek.....2
3. Build source code.....2
4. Flash images.....3

1. Preparation

- Ubuntu 22.04
- Install repo tool

  mkdir -p ~/bin
  cd ~/bin
  wget http://storage.googleapis.com/git-repo-downloads/repo
  chmod +755 repo
  export PATH=~/bin:$PATH
- Install required packages

  sudo apt update
  sudo apt-get install lz4 zstd diffstat qemu-user-static xmlstarlet texinfo chrpath gcc-aarch64-linux-gnu libarchive-dev ssh libselinux1-dev fakechroot g++-aarch64-linux-gnu \
  libiberty-dev qemu-user-static qemu-user g++ gawk gcc make libwayland-dev fakeroot libpam0g-dev binutils-dev util-linux uuid-dev liblz4-tool libxml-simple-perl libxml2-utils \
  libc++-dev libstdc++-12-dev m4 unzip cpio wget build-essential zstd python3-pip libc6-i386 lib32stdc++6 binfmt-support gcc-arm-none-eabi binutils-arm-none-eabi kconfig-frontends \
  python3-rosdep2 libgtest-dev python3-lxml libncurses5 scons qemu-user liblz4-dev libcurl4-openssl-dev

  sudo rm -rf /lib/ld-linux-aarch64.so.1
  sudo ln -sf /usr/aarch64-linux-gnu/lib/ld-linux-aarch64.so.1 /lib/ld-linux-aarch64.so.1
  sudo ln -sf /bin/bash /bin/sh

wget
http://cz.archive.ubuntu.com/ubuntu/pool/universe/f/fakechroot/libfakechroot_2.20.1%2Bds-15_amd64.deb
sudo dpkg -i ./libfakechroot_2.20.1+ds-15_amd64.deb
wget
http://cz.archive.ubuntu.com/ubuntu/pool/universe/f/fakechroot/fakechroot_2.20.1%2Bds-15_all.deb
sudo dpkg -i ./fakechroot_2.20.1+ds-15_all.deb

sudo perl -MCPAN -e 'install XML::Simple'

pip3 install dtschema==2021.10
pip3 install jsonschema==4.0.0
pip3 install ruamel.yaml==0.17.17
pip3 install requests

2. Get source code from HSPTek

- Check and send to HSPTek engineer ssh public key “cat ~/.ssh/id_rsa.pub” (Use “ssh-keygen” to generate if not exist)
- Download source code:
  mkdir ~/hlos # change to yours
  cd ~/hlos
  repo init -u ssh://customer@git.hsptek.com:22922/QCS8550_UBUNTU/manifests -b ubuntu_r00015.1
  mkdir .repo/local_manifests
  cp .repo/manifests/kernel.xml .repo/manifests/framework.xml .repo/local_manifests/
  repo sync -c -j8

3. Build source code

3.1. Build kernel

cd hlos/src/kernel-5.15/kernel_platform
export BUILD_CONFIG=msm-kernel/build.config.msm.kalama.lu VARIANT=consolidate OUT_DIR=../out/msm-kernel-kalama_lu-consolidate
./build/build.sh -j8

3.2 Build system

```
cd hlos
export MACHINE=kalama DISTRO=qti-distro-ubuntu-fullstack-debug
source poky/qti-conf/set_bb_env.sh
bitbake qti-ubuntu-robotics-image
```

4. Flash images

- After build complete, HLOS images will be at ~/hlos/build-qti-distro-ubuntu-fullstack-debug/tmp-glibc/deploy/images/kalama/qti-ubuntu-robotics-image/
- Boot box into fastboot mode by command “adb reboot bootloader" via Type-C cable or "systemctl reboot bootloader" by ssh , then check “fastboot devices"
- Use bellow commands to flash hlos images into box:

```
cd
~/hlos/build-qti-distro-ubuntu-fullstack-debug/tmp-glibc/deploy/images/kalama/qti-ubuntu-robotics-image/
fastboot flash abl_a abl.elf
fastboot flash abl_b abl.elf
fastboot flash dtbo_a dtbo.img
fastboot flash dtbo_b dtbo.img
fastboot flash boot_a boot.img
fastboot flash boot_b boot.img
fastboot flash system system.img
fastboot reboot
```